VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024 — Surpassing DAPO by 10+ Points

ByteDance Seed

Base Model: Qwen2.5-32B | No SFT Data

AIME 2024 avg@32: 60.4 | 5,000 Training Steps | Zero Crashes

arXiv:2504.05118

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-Model-Free (GRPO, DAPO)

• Trajectory-level advantage assigned uniformly to all tokens

• High variance in credit estimation (Monte Carlo)

• No explicit state-value estimation during generation

• Struggles with subtle step-level errors that cause failure

Value-Model-Based (VAPO)

• Token-level credit assignment tracing each action's impact

• Lower-variance value estimates per token

• Generalizes across samples, improving sample efficiency

• Critical when long CoT reasoning requires precise step evaluation

Insight: For long CoT, uniform trajectory-level credit is too coarse. Value models enable fine-grained, token-level evaluation essential for multi-step reasoning.

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

1. Value Model Bias

Long trajectories + bootstrapped learning amplify bias in value estimates, especially early in training.

2. Heterogeneous Sequence Lengths

Short and long responses require different bias-variance tradeoffs; a single λ is suboptimal.

3. Reward Signal Sparsity

Only the final <eos> token receives reward, making credit assignment across thousands of tokens extremely hard.

These challenges explain why value-model-based RL has been largely abandoned in favor of value-free methods like GRPO — until now.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Bias

Value-Pretraining

Pre-trains value model before RL to reduce initial bias

Decoupled-GAE

Separates GAE computation for robust value estimation

Heterogeneous Lengths

Length-Adaptive GAE

Dynamic λ adjustment: short→low λ, long→high λ

Sparsity / General

Clip-Higher

Asymmetric clipping encourages exploration

Token-level Loss

Token-level normalization vs sequence-level

Positive Example LM Loss

Self-imitation on correct trajectories

Group-Sampling

Multiple responses per prompt for better baseline

All components jointly address the three challenges. No single technique is sufficient — the integration is what enables stable, high-performance value-model-based RL.

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

Core Idea

- GAE λ controls bias-variance tradeoff:
  - Low λ → low variance, high bias (good for short sequences)
  - High λ → low bias, high variance (good for long sequences)
- Fixed λ is suboptimal when response lengths vary wildly
- Length-Adaptive GAE dynamically selects λ based on response length

Short Response

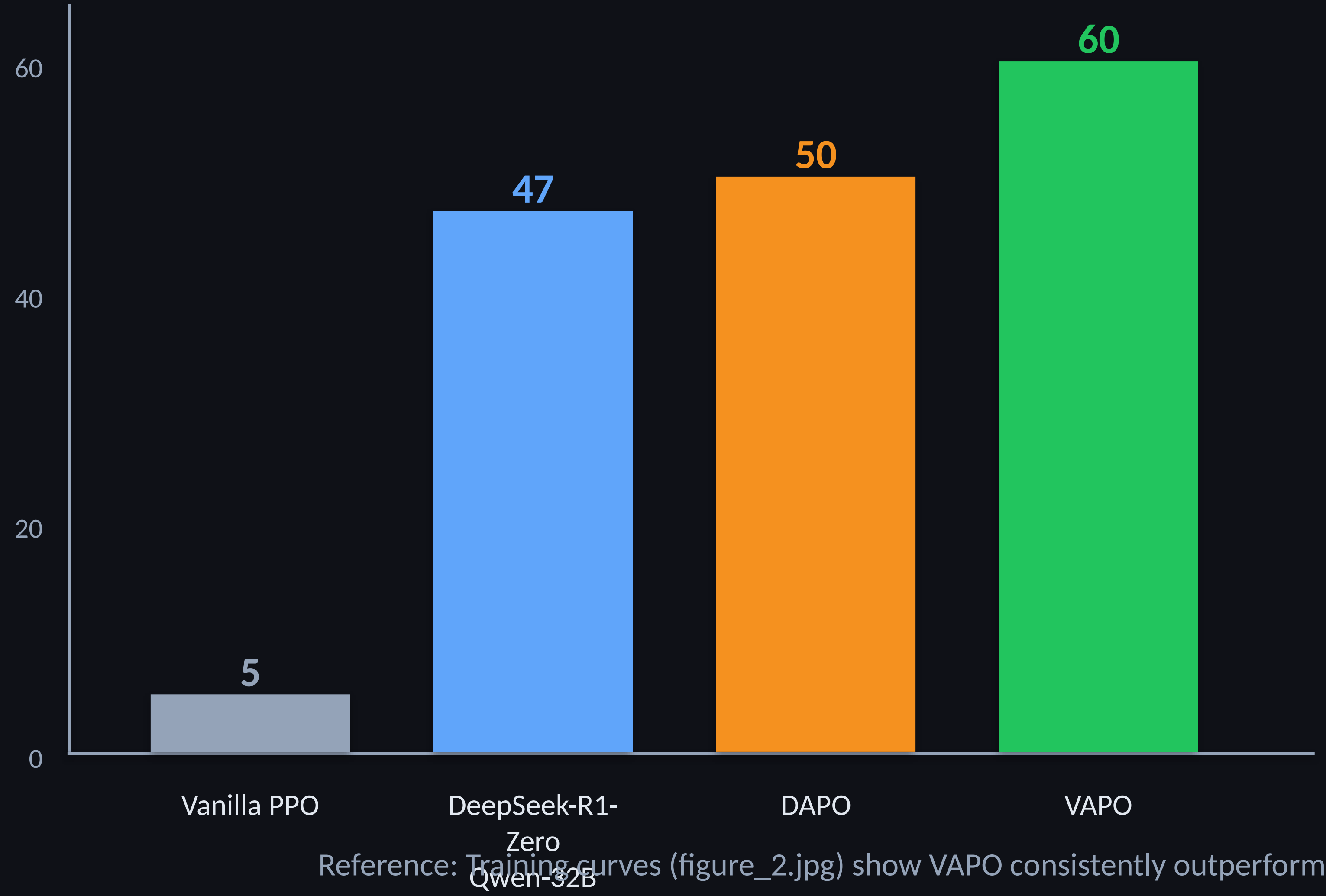
Lower λ → lower variance
Reliable value estimates with fewer tokens
Less noise in credit assignment

Long Response

Higher λ → lower bias
Accurate credit propagation across many steps
Reduces systematic underestimation of returns

Impact: Removing Length-Adaptive GAE drops AIME score from 60 → 45 (15-point contribution). It is a novel technique addressing a problem unique to long-CoT reasoning.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



Key Results

- VAPO: 60.4 AIME 2024 avg@32
- DAPO: 50 (+10.4 gap)
- DeepSeek-R1-Zero: 47 (+13.4 gap)
- Vanilla PPO: 5 (baseline)
- Same base model (Qwen2.5-32B)
- No supervised fine-tuning data

Reference: Training curves (figure_2.jpg) show VAPO consistently outperforms DAPO throughout training.

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

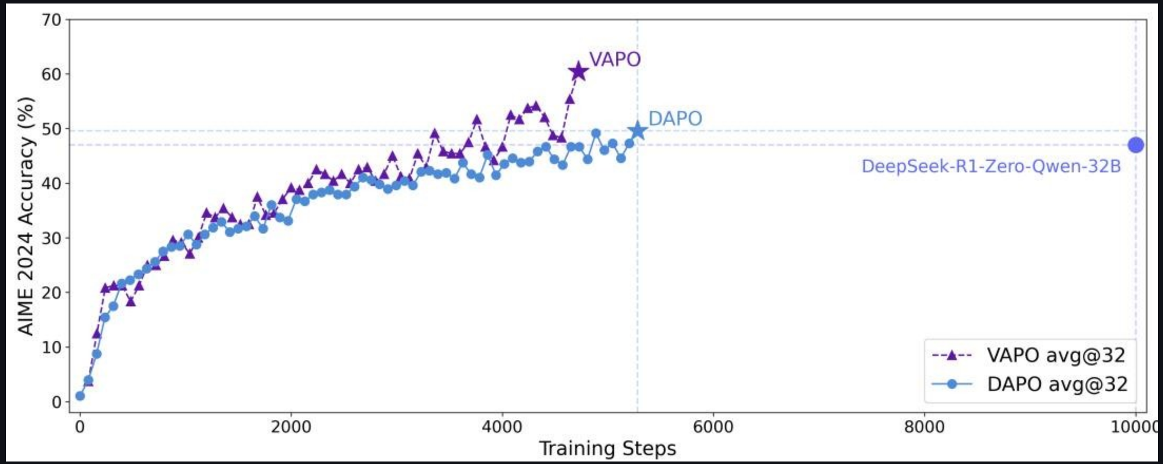


Figure 2: VAPO vs DAPO training curves on AIME 2024

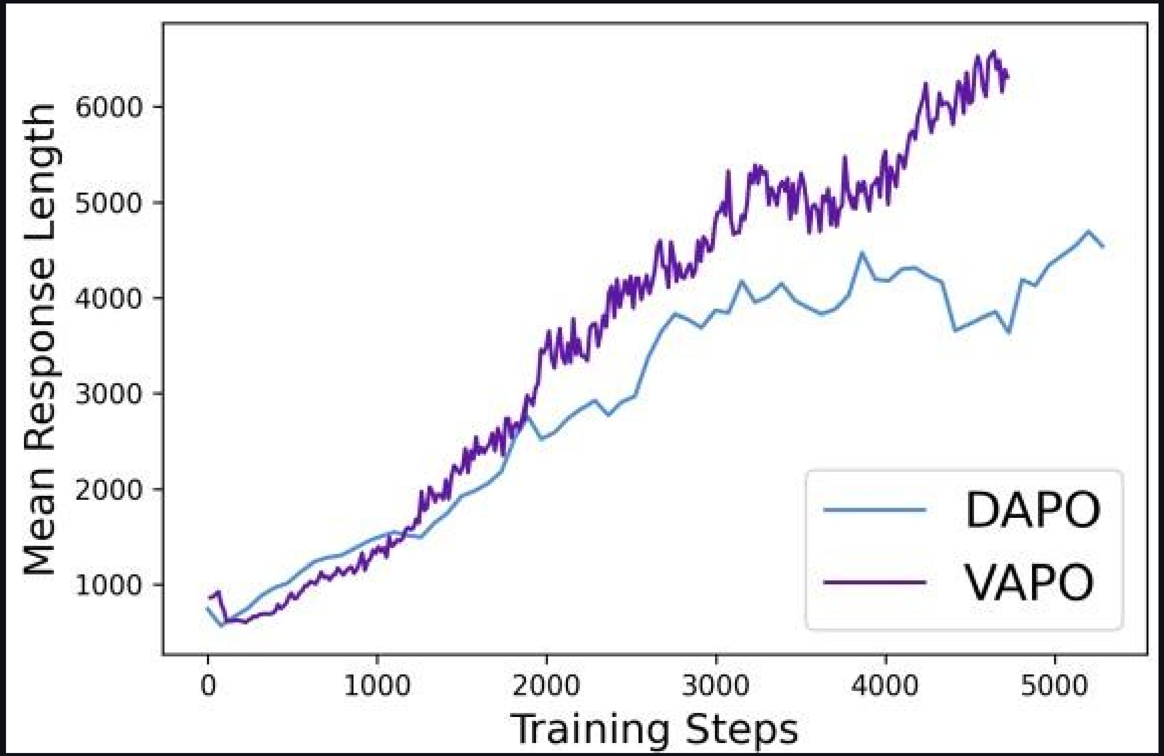


Figure 3: Mean response length over training

Accuracy: VAPO reaches ~60% at step 4,500
DAPO plateaus near 50% at step 5,500
DeepSeek-R1-Zero reaches 47% after ~10,000 steps

Response Length: VAPO generates longer chains (~6,000 tokens) vs DAPO (~4,500 tokens), indicating more elaborate reasoning
Zero training crashes across multiple independent runs

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	AIME24 Score	
VAPO (full)	60	
w/o Positive Example LM Loss	54	
w/o Token-level Loss	53	
w/o Group-Sampling	55	
w/o Clip-Higher	46	
w/o Length-Adaptive GAE	45	-15 pts
w/o Decoupled-GAE	33	-27 pts
w/o Value-Pretraining	11	-49 pts

Value-Pretraining is the single most critical component: without it, value-model-based RL barely works (11 vs 60). Decoupled-GAE and Length-Adaptive GAE are the next most impactful.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1

Value-model-based RL can surpass value-free methods

When bias, heterogeneous lengths, and sparsity are systematically addressed, value models deliver finer credit assignment and lower variance — translating to +10 points over DAPO.

2

Length-Adaptive GAE is critical for long-CoT reasoning

A novel technique addressing a unique problem: dynamic λ adjustment contributes 15 points and enables optimal bias-variance tradeoffs across variable sequence lengths.

3

Value-Pretraining is the single most important component

Removing it drops the score from 60 to 11. Pre-training the value model before RL is non-negotiable for stable value-based training.

4

VAPO is highly training-efficient and stable

Reaches SOTA in ~5,000 steps with zero training crashes across multiple runs — outperforming DAPO in both speed and final accuracy.

5

Implications for future reasoning models

Value-model-based RL should be reconsidered as a first-class approach for complex reasoning. The VAPO framework provides a principled recipe for overcoming its historical limitations.

VAPO demonstrates that with careful engineering, value-model-based RL achieves state-of-the-art math reasoning — setting a new benchmark for future research.